AQUA SPHERE

AI Dashboard

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Smart Water Management Platform

for Bengaluru

Connecting citizens, tanker operators, and BWSSB to eliminate water scarcity, price gouging, and distribution inequality.

Hackathon 2026

BMS College of Engineering

A Section

01 / PROBLEM

Why does Bengaluru need AquaSphere?

Bengaluru faces acute seasonal water scarcity. Citizens depend on private tankers but face price gouging, no delivery transparency, and zero visibility into local water levels.

4.2M+
people affected

60%
tanker price hike

3 hrs/day
average shortage

01

Price Gouging

Tanker rates spike 60% or more during summer — no regulation, no accountability.

02

Uneven Supply

Wealthy wards hoard supply. Low-income areas run dry with no recourse.

03

Zero Visibility

Citizens can't track deliveries, report leaks, or see reservoir levels.

Three modules. One unified platform.

AquaSphere connects citizens, tanker operators, and BWSSB officials through an integrated React + Firebase web platform.

Citizen App

For residents

- Request water delivery
- Report pipe leaks
- Live reservoir levels
- Track delivery status

Tanker Routing

For operators

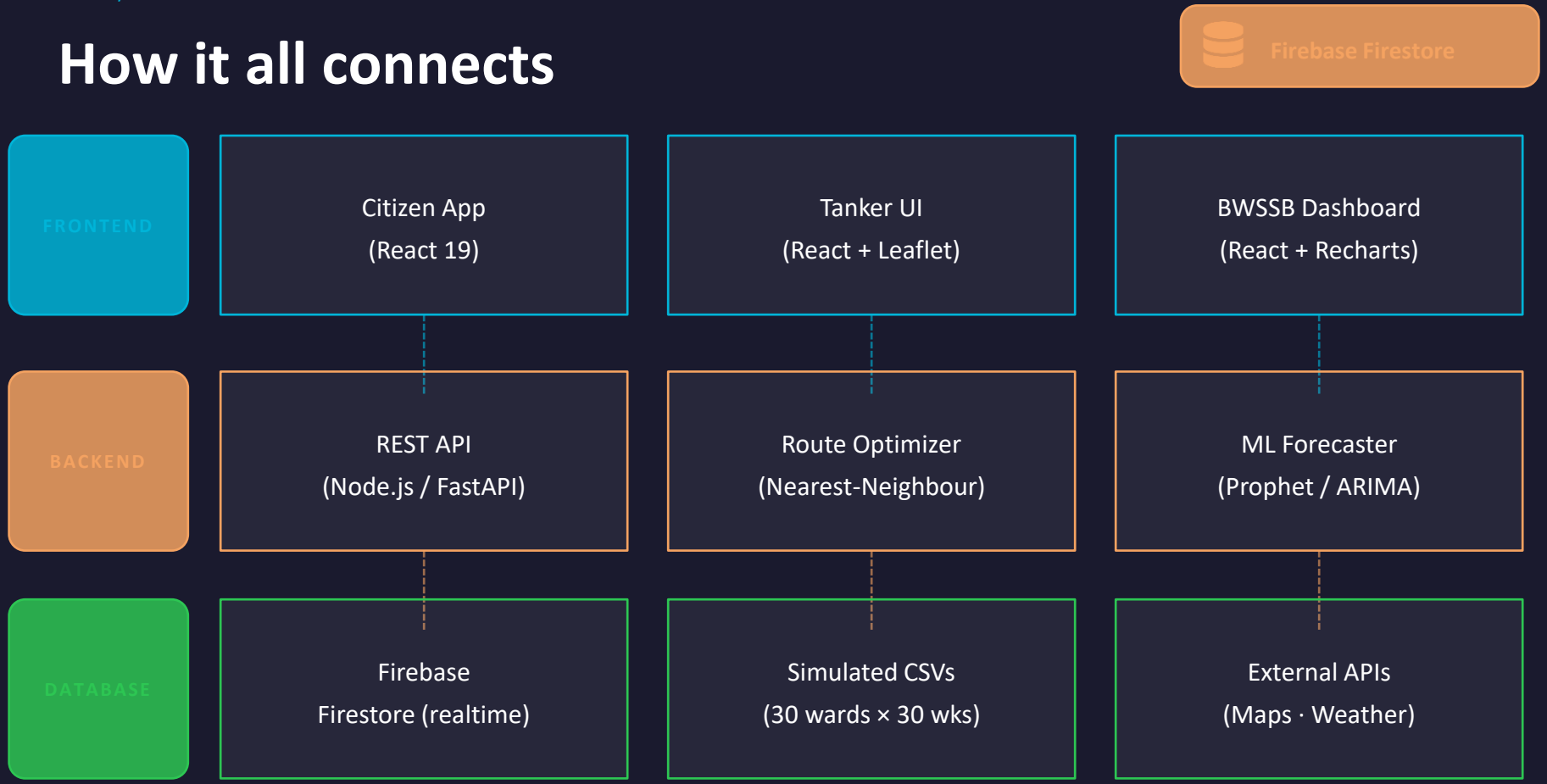
- Optimized route algorithm
- Urgency-based dispatch
- GPS live tracking
- Fuel cost estimator

Predictive Dashboard

For BWSSB

- 7-day shortage forecast
- Ward-level risk heatmap
- ML-powered alerts
- Historical trends

How it all connects



What we built it with

Frontend

- React 19 + TypeScript
- Vite 7 (build tool)
- Tailwind CSS 4
- Framer Motion
- Recharts (charts)
- Leaflet + React-Leaflet

Backend & ML

- Node.js / FastAPI
- Facebook Prophet
- ARIMA time-series
- Nearest-Neighbour routing
- OSRM open-source routing
- Python Faker (sim data)

Database & Tools



Firebase Firestore

Realtime NoSQL database

› Google Maps JS API

› OpenWeather API

› Git & GitHub

› npm · Node.js v18+

› Render / Railway (deploy)

What makes AquaSphere work

Multi-Role Access

Citizen, Driver, Admin & Manager views with role-based Firebase auth

Live Map (Leaflet)

Interactive ward map with real-time tanker & leak markers

AI Shortage Forecast

Prophet ML model predicts ward-level shortages 7 days ahead

Incident Monitoring

Citizens report leaks with severity; BWSSB gets instant alerts

Route Optimization

Nearest-neighbour algorithm dispatches tankers by urgency & distance

Firebase Realtime DB

Firestore syncs requests, tanker locations & forecasts live

Future Scope

IoT Sensors

Live groundwater & pipe pressure from field sensors, replacing simulated CSV data

Flutter Mobile App

Native iOS/Android app with offline support and push notifications for citizens

LSTM Deep Learning

Advanced models trained on real BWSSB data for hyper-local ward forecasting

Sustainability

Rainwater harvesting recommendations + groundwater recharge zone mapping per ward